BEING A LOCALLY RINGED SITE IS INTRINSIC, I.E. INVARIANT UNDER MORPHISMS AND EQUIVALENCES OF TOPOI

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites one related lemma and an equation, recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Equivalent characterization of the locally ringed condition. Stacks Project, Modules on Sites, Lemma lemma-locally-ringed (tag 09BJ). Supplies the isomorphism and surjection whose pullback is used to conclude.
- Exactness properties of the inverse image functor. Stacks Project, Sites, section on morphisms of topoi. Justifies that the relevant limits, colimits and (iso)morphisms are preserved.

Source reference. Original equation 05D7: sources/stacks-chow/tags.txt, source lines 6945–6945. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 04ES: sources/stacks-chow/tags.txt, source lines 5773–5773. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Being a locally ringed site is an intrinsic property. More precisely,*

(1) *if $f : Sh(\mathcal{C}') \to Sh(\mathcal{C})$ is a morphism of topoi and $(\mathcal{C}, \mathcal{O})$ is a locally ringed site, then $(\mathcal{C}', f^{-1}\mathcal{O})$ is a locally ringed site, and*
(2) *if $(f, f^\sharp) : (Sh(\mathcal{C}'), \mathcal{O}') \to (Sh(\mathcal{C}), \mathcal{O})$ is an equivalence of ringed topoi, then $(\mathcal{C}, \mathcal{O})$ is locally ringed if and only if $(\mathcal{C}', \mathcal{O}')$ is locally ringed.*

Proof. It is clear that (2) follows from (1). To prove (1) note that as f^{-1} is exact we have $f^{-1}* = *$, $f^{-1}\emptyset^\# = \emptyset^\#$, and f^{-1} commutes with products, equalizers and transforms isomorphisms and surjections into isomorphisms and surjections. Thus f^{-1} transforms the isomorphism (05D7) into its analogue for $f^{-1}\mathcal{O}$ and transforms the surjection of Lemma 04ES part (3) into the corresponding surjection for $f^{-1}\mathcal{O}$. □

Date: stacks-project:sites-modules.tex:local-snapshot-20261012; GNU FDL 1.2 or later.